

1. Verificación de que systemd es el init

```
-bash-5.3# ps -p 1 -o pid,comm,args | tee "$EVD/1-pid1.txt"
  PID COMMAND          COMMAND
    1 systemd          /sbin/init
-bash-5.3#
```

```
-bash-5.3# pstree -p 1
systemd(1)─bash(241)─pstree(432)
           │
           ├──dbus-daemon(240)
           ├──sshd(242)
           ├──systemd-journal(123)
           ├──systemd-logind(243)
           ├──systemd-network(195)
           ├──systemd-nsresou(156)─systemd-nsresou(419)
           │                     ├──systemd-nsresou(420)
           │                     ├──systemd-nsresou(421)
           │                     ├──systemd-nsresou(422)
           │                     └──systemd-nsresou(423)
           ├──systemd-oomd(129)
           ├──systemd-resolve(159)
           ├──systemd-timesyn(183)─{systemd-timesyn}(187)
           └──systemd-udev(184)
```

2. Verificación de que systemd está activo

```
-bash-5.3# systemctl is-system-running
running
-bash-5.3#
```

```

├─464 systemctl status --no-pager
├─465 tee /root/evidencias-systemd/2-status.txt
├─systemd-journald.service
│   └─123 /usr/lib/systemd/systemd-journald
├─systemd-logind.service
│   └─243 /usr/lib/systemd/systemd-logind
├─systemd-networkd.service
│   └─195 /usr/lib/systemd/systemd-networkd
├─systemd-nsresourced.service
│   ├──156 /usr/lib/systemd/systemd-nsresourced
│   ├──457 "systemd-nsresourcework: waiting..."
│   ├──458 "systemd-nsresourcework: waiting..."
│   ├──459 "systemd-nsresourcework: waiting..."
│   ├──460 "systemd-nsresourcework: waiting..."
│   └─461 "systemd-nsresourcework: waiting..."
├─systemd-oomd.service
│   └─129 /usr/lib/systemd/systemd-oomd
├─systemd-resolved.service
│   └─159 /usr/lib/systemd/systemd-resolved
├─systemd-timesyncd.service
│   └─183 /usr/lib/systemd/systemd-timesyncd
├─systemd-udev.service
│   ├──udev
│   └─184 /usr/lib/systemd/systemd-udev
-bash-5.3#
```

3. Servicio real con systemd

```
-bash-5.3# ssh-keygen -A
```

```
-bash-5.3# ssh-keygen -A
-bash-5.3# systemctl enable --now sshd
-bash: systemctl: command not found
-bash-5.3# systemctl enable --now sshd
> systemctl enable --now sshd
-bash-5.3# systemctl status sshd --no-pager
• sshd.service - OpenSSH Daemon
   Loaded: loaded (/etc/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 23:45:00 -03; 1h 14min ago
 Invocation: e67294dfe00449cd88f185e6427619b0
   Docs: man:sshd(8)
        man::sshd_config(5)
  Main PID: 242 (sshd)
    Tasks: 1 (limit: 9533)
   Memory: 2.1M (peak: 2.6M)
      CPU: 43ms
   CGroup: /system.slice/ssh.service
           └─242 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 28 23:45:00 lfs systemd[1]: Starting OpenSSH Daemon...
Jan 28 23:45:00 lfs sshd[242]: Server listening on 0.0.0.0 port 22.
Jan 28 23:45:00 lfs sshd[242]: Server listening on :: port 22.
Jan 28 23:45:00 lfs systemd[1]: Started OpenSSH Daemon.
-bash-5.3#
```

```
-bash-5.3# ss -tulpn | grep ':22'
tcp    LISTEN 0      0      0.0.0.0:22      0.0.0.0:*      users:(("sshd",p
id=242,fd=6))
tcp    LISTEN 0      0      *:22          *:.*          users:(("sshd",p
id=242,fd=7))
-bash-5.3#
```

```
-bash-5.3# ssh root@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:u27AtFuz5HprFcIj17oxf47oL+yQPTPtQgcq5r8yDmY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
root@localhost's password:
Last login: Thu Jan 15 23:02:42 2026 from 10.0.2.2
-bash-5.3#
```

4. Custodiar logs o capturas de configuración de unidades y servicios

```
-bash-5.3# systemctl cat sshd
# /etc/systemd/system/ssh.service
[Unit]
Description=OpenSSH Daemon
Documentation=man:sshd(8) man::sshd_config(5)
After=network.target
[Service]
Type=notify
ExecStart=/usr/sbin/sshd -D
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartSec=3
[Install]
WantedBy=multi-user.target
-bash-5.3# _
```

```
-bash-5.3# journalctl -u sshd -n 100 --no-pager
```

```
Jan 28 13:40:58 lfs systemd[1]: Starting OpenSSH Daemon...
Jan 28 13:40:58 lfs sshd[243]: Server listening on 0.0.0.0 port 22.
Jan 28 13:40:58 lfs sshd[243]: Server listening on :: port 22.
Jan 28 13:40:58 lfs systemd[1]: Started OpenSSH Daemon.
Jan 28 13:41:25 lfs systemd[1]: Stopping OpenSSH Daemon...
Jan 28 13:41:25 lfs sshd[243]: Received signal 15; terminating.
Jan 28 13:41:25 lfs systemd[1]: sshd.service: Deactivated successfully.
Jan 28 13:41:25 lfs systemd[1]: Stopped OpenSSH Daemon.
-- Boot 22607e14ac9e49acb02bff00decc5c1a --
Jan 28 22:11:37 lfs systemd[1]: Starting OpenSSH Daemon...
Jan 28 22:11:37 lfs sshd[242]: Server listening on 0.0.0.0 port 22.
Jan 28 22:11:37 lfs sshd[242]: Server listening on :: port 22.
Jan 28 22:11:37 lfs systemd[1]: Started OpenSSH Daemon.
-- Boot be1903d79f7247dd90d5b1a2d50d9c4b --
Jan 28 23:45:00 lfs systemd[1]: Starting OpenSSH Daemon...
Jan 28 23:45:00 lfs sshd[242]: Server listening on 0.0.0.0 port 22.
Jan 28 23:45:00 lfs sshd[242]: Server listening on :: port 22.
Jan 28 23:45:00 lfs systemd[1]: Started OpenSSH Daemon.
Jan 29 01:04:52 lfs sshd-session[550]: Accepted password for root from ::1 port
34808 ssh2
Jan 29 01:07:30 lfs sshd-session[552]: Received disconnect from ::1 port 34808:1
1: disconnected by user
Jan 29 01:07:30 lfs sshd-session[552]: Disconnected from user root ::1 port 3480
8
-bash-5.3# _
```

5. Reflexión sobre los beneficios de systemd frente a métodos antiguos de init

- systemd gestiona servicios con unidades y dependencias (más ordenado que scripts rc).
- Permite habilitar/arrancar/parar/ver estado con un comando (systemctl).
- Soporta arranque paralelo y mejora tiempos de inicio.
- journald centraliza logs y permite filtrar por servicio (journalctl -u sshd).
- Da control y supervisión (reinicios, límites, estados) de forma estándar.

